Car Price Prediction Based on Multiple Machine Learning Models

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Abstract: This article takes car price prediction based on three machine learning models as the topic. Cars, especially in recent decades, are becoming increasingly necessary for companies and families around the world. As the demand for cars grows, further concerns about purchasing vehicles emerge. Obviously, car price acts as a key factor in making these decisions. Therefore, car price prediction becomes a meaningful topic to discuss about. In this passage, comparisons between three machine learning models, Linear Regression, Random Forest and XGBoost, are carried out. The three models are applied to a train and a test dataset and the performances are evaluated by Root Mean Squared Error (RMSE) and accuracy. For the train dataset, the RMSE for Linear Regression, Random Forest and XGBoost are 534,865.15, 155,706.68 and 302,861.88 while the accuracy reveals to be 61.64%, 96.75% and 87.70%. For the test dataset, the RMSE for Linear Regression, Random Forest and XGBoost are 555,802.12, 338,065.9 and 337,698.16 while the accuracy is 58.44%, 84.62% and 84.66%. The overall conclusion is that Random Forest learnt slightly faster and better than XGBoost but performed almost the same in predicting car prices. Linear regression performed the worst throughout all datasets.

1 INTRODUCTION

Humans have taken advantage of an immense amount of means of transportation. One category of popular travel tools is cars. Invented in 1886, cars are now becoming an increasingly indispensable part of people's life. More companies and families are owning vehicles of different types recently, many of which own two or more. Cars are often regarded as dependable tools during work period and are considered as reliable friends for travellers. From application scenarios of cars, it is easy to notice that the mobilization efficiency has been greatly increased and the range of human activities on land has been greatly expanded. In recent years, the demand for cars has met a substantial surge. When considering purchasing a car, the price of it serves as a vital factor, while it could differ greatly due to a variety of aspects. After the pandemic and experiencing dramatic changes in energy resources, it has been widely acknowledged that wide swings can be common in the car market over the near future. Therefore, more advanced methods are required to be applied to predict car prices, which can contribute to making vehicle purchase decisions. Furthermore, this will also provide vehicle companies with convenience in pricing their automobiles and defining future development direction.

With the popularization of Artificial Intelligence (AI) and relevant technologies (Roll, 2016; Chen, 2022; Patil, 1998), it is apparent that algorithms are playing vital roles on various occasions. In recent years, lots of representative algorithms have appeared, including the random forest, decision tree, logistic regression etc. Machine learning algorithms have been widely applied to various fields, such as chemistry, biomedicine and especially business analytics. For instance, Diamantaras et al. used eight machine learning models to predict airfare prices (Diamantaras, 2017). Karthikeyan et al. used ensemble-based machine learning techniques to carry out gold price prediction (Karthikeyan, 2019). Michael et al. determined how variations in the Deoxyribonucleic Acid (DNA) of individuals can affect the risk of different diseases through machine learning (Alipanahi, 2015). One important field included is car price prediction. For instance, Gegic et al. used artificial neural network, support vector

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machine and random forest to predict car price (Gegic, 2019). It is now a fact that AI can bring about accurate prediction results. Therefore, this paper intends to consider using machine learning models to predict the corresponding price according to various features of cars and analyze the importance of the contribution factors.

The dataset used in the study is taken from Kaggle (Kaggle, 2021). The study considers using three machine learning models to predict car prices, including Linear Regression, Random Forest and XGBoost. The implementation of code was referred to a notebook written on Kaggle (Kaggle, 2019). Standards used to compare their performances are Root Mean Squared Error (RMSE) and accuracy. Feature importance function in Random Forest was also used to evaluate the importance of the contribution factors.

2 METHOD

2.1 Dataset Preparation

The dataset used in this study is taken from Kaggle (Kaggle, 2019). The original dataset includes information of 6, 1500 vehicles and 22 variables. Before inputting the dataset, 11 of them such as 'url', 'city' was dropped due to the existence of NaN columns. In addition, 'Year' and 'Odometer' features were transferred into int type.

Before inputting the dataset, it was split into a train dataset and a test dataset. The train dataset included 525, 839 rows and the test dataset included 34, 440 rows.

2.2 Machine Learning Models-based Prediction

This study used Linear Regression, Random Forest and XGBoost for car price prediction. The models were imported from sklearn. The evaluation metrics applied in this study included RMSE and accuracy.

2.2.1 Linear Regression

Linear Regression is a widely utilized machine learning model, commonly employed in statistical analysis to explore and predict relationships between variables. It operates on the principle of fitting a straight line to a set of data points in a manner that best captures the underlying relationship between two features, typically used for continuous target variables.

In practice, Linear Regression analyzes paired data to determine a direct correlation, making it especially valuable in scenarios where the relationship between the independent variable (or predictor) and the dependent variable (or outcome) is expected to be linear. This model constructs a linear equation based on the input dataset, plotting a line in a two-dimensional coordinate system aimed at best fitting the observed data points. This linear representation enables predictions regarding future outcomes by projecting the dependent variable's values based on current trends.

However, the real-world application of Linear Regression acknowledges that the model cannot perfectly fit every data point. There will invariably be deviations between the actual values and the values predicted by the model, which are termed as residuals. The presence of these residuals indicates the error inherent in the model predictions, reflecting the nuances and variability in real-world data that cannot be fully captured by a linear model.

To evaluate the performance of a Linear Regression model, a metric known as Root Mean Square Error (RMSE) is used. RMSE measures the average magnitude of the residuals, providing insight into how much error is typically involved in the predictions made by the linear model. A lower RMSE value indicates a better fit of the model to the data, suggesting that the linear equation is effective at capturing the primary trends and reducing the prediction errors.

Despite its simplicity, Linear Regression has profound implications across various fields, from economics and finance to healthcare and social sciences. It aids in understanding and quantifying the strength of relationships between variables, facilitating decision-making processes by providing a clear, quantifiable link between different variables. For instance, businesses often use Linear Regression to predict sales based on advertising spend, or healthcare providers might use it to predict patient outcomes based on treatment protocols.

2.2.2 Random Forest

Random Forest is an ensemble learning method that builds on the simplicity of decision trees by combining multiple such trees to form a more robust and accurate model. This technique leverages the strength of numerous decision trees, each serving as an estimator, to improve predictive accuracy and control over-fitting, which is a common problem in single decision tree models. The core idea behind

Random Forest is to create a 'forest' of trees where each tree is slightly different from the others.

The model begins by using a technique called Bootstrap Aggregating, or bagging, to create different subsets of the original dataset, with replacement. Each subset is used to train a separate decision tree. Since the subsets can have overlapping data points, each tree in the forest is not identically trained, which helps in making the ensemble of trees less sensitive to the specifics of the training data, thereby enhancing the generalization capabilities of the model.

During prediction, Random Forest takes an average of the outputs from all the decision trees for regression problems, or a majority vote for classification, to improve predictive accuracy. By aggregating the results, the model not only reduces the variance but also combines the individual trees' predictions to produce a more accurate and stable estimation than any single tree could provide.

Despite its many benefits, such as high accuracy, the ability to handle a large number of input variables, and robustness to overfitting, the Random Forest model does have some drawbacks. One significant limitation is the loss of interpretability that comes with using a single decision tree. The complex structure of multiple trees makes it difficult to visualize and understand the specific reasons behind predictions, thus complicating the explanation of model decisions to stakeholders who are not technical experts.

Furthermore, while Random Forest is generally good at handling overfitting, it can still perform poorly in the presence of noisy classification or regression problems. This occurs because noise can lead to overly complex trees, which, even when averaged, result in an overfitted model. Therefore, while Random Forest is a powerful tool in the arsenal of machine learning techniques, it requires careful tuning of parameters like the number of trees in the forest and the depth of each tree to optimize its performance and prevent overfitting. This ensemble method, hence, offers a balance between accuracy and complexity, providing a reliable predictive tool across various applications in fields ranging from finance to healthcare.

2.2.3 XGBoost

XGBoost is an abbreviation for eXtreme Gradient Boosting. It is a machine learning model with massively parallel boosting trees. The whole process is an additive model made up by multiple base models. When generating a processing tree, algorithms including greedy algorithm for best solutions and approximation algorithm for massive data, can be practical alternatives. Advantages for the XGBoost are high accuracy and flexibility, learning shrinkage, regularization included, column sampling and various tools to support parallel. Disadvantages also exist, during the whole process, all data have to be traversed. Furthermore, the prearrangement process requires relatively high space complexity to store the index of the gradient statistics for the corresponding sample.

3 RESULTS AND DISCUSSION

The results of the predictions are shown in Table 1. Illustrated by the Table, it's obvious that for the train dataset, Random Forest reached the highest accuracy of 96.75% while XGBoost demonstrated fairly well, achieving 87.70%, followed by Linear Regression which appeared to be the worst among the three with an accuracy of 61.64%. The result is almost the same when it comes to train RMSE. Random Forest, the best among all three, had the lowest RMSE of 155,706.68, next is XGBoost with the value of 302,861.88 and Linear Regression with the highest RMSE of 534,865.15.

Table 1. The performance of different models

Model name	Train Accuracy	Train RMSE	Test Accuracy	Test RMSE
Linear Regression	61.64%	534,865.15	58.44%	555,802.12
Random Forest	96.75%	155,706.68	84.62%	338,065.9
XGBoost	87.70%	302,861.88	84.66%	337,698.16

However, while applying them to the test dataset, the accuracy of Random Forest was almost the same as XGBoost which were both around 85%, 84.62% and 84.66% representatively, while Linear regression kept carrying out the worst performance with an accuracy of 58.44%. The same condition emerged when considering RMSE. The RMSE of Random Forest and XGBoost were about the same, 338,065.9 and 337,698.16 each, and the largest one remained Linear Regression with a value of 555,802.12, almost the same as the train RMSE.

Above all, Random Forest learnt slightly faster and better than XGBoost as Random Forest performed evidently better when greeting the train dataset. However, Random Forest predicted as accurately as XGBoost did according to the test dataset results. It's obvious to notice that Linear Regression performed the worst throughout all datasets.

It was a possible reason that Linear regression is a one-time analyzation while others went through two times or more. What's more, multiple results were generated and concluded in both Random Forest and XGBoost, while only one was generated from Linear Regression.

Noticeably, some of the reasons concluded are only based on the understanding towards the algorithms and are lack of further research data support. However, in a study written by Vishal Khandare and Manish Pandey in which they used the same three models to predict electric car prices, the results come out to be almost the same (Khandare, 2022). In the studies mentioned above, Random Forest performs the best, reaching an accuracy of 90.38% while XGBoost gets 89% and Linear Regression achieves 67%. Therefore, the result is quite credible to a certain extent. However, the reasons and logic behind it still need further work and research.

4 CONCLUSIONS

This paper compared the performances between three machine learning models which are Linear Regression, Random Forest and XGBoost representatively. The standards used to appraise the results are Root Mean Squared Error (RMSE) and accuracy. Through the final results, it is evident that Random Forest learnt slightly faster and better than XGBoost but performed almost the same in predicting car prices. Linear regression performed the worst throughout all datasets. It was a possible reason that Linear regression is a one-time analyzation and only one result is concluded, much less than the other two models.

This article supports the feasibility of machine learning models and analyzed the reasons why the performances differed from each other. The reasons mentioned above are according to the personal perspectives and understanding and requires more accurate data to support them. However, these can still serve as a clue to analyze different machine learning models and select the best to be applied to car price prediction.

REFERENCES

Alipanahi, B., Leung, M. K., Delong, A., & Frey, B. J. 2015. Machine learning in genomic medicine: a review of computational problems and data sets. Proceedings of the IEEE, 104(1), 176-197.

Chen, X., Zou, D., Xie, H., Cheng, G., & Liu, C. 2022. Two decades of artificial intelligence in education. Educational Technology & Society, 25(1), 28-47.

Diamantaras, K. I., Tziridis, K., Kalampokas, T., & Papakostas, G. A. 2017. Airfare prices prediction using machine learning techniques. 2017 25th European Signal Processing Conference (EUSIPCO), 1036-1039. IEEE.

Gegic, E., Isakovic, B., Keco, D., Masetic, Z., & Kevric, J. 2019. Car price prediction using machine learning techniques. TEM Journal, 8(1), 113.

Kaggle. 2019. Used cars price prediction by 15 models. Retrieved from https://www.kaggle.com/code/vbmokin/used-cars-price-prediction-by-15-models/notebook

Kaggle. 2021. Craigslist carstrucks data. Retrieved from https://www.kaggle.com/datasets/austinreese/craigslist-carstrucks-data

Karthikeyan, P., & Manjula, K. A. 2019. Gold price prediction using ensemble based machine learning techniques. 2019 3rd International Conference on Trends in Electronics and Informatics (ICOEI), 1360-1364. IEEE.

Khandare, V., & Pandey, M. 2022. Electric Car Price Prediction using Machine Learning Techniques. International Journal of Engineering Research in Computer Science and Engineering (IJERCSE), 9(11).

Patil, R. S., Szolovits, P., & Schwartz, W. B. 1988. Artificial intelligence in medical diagnosis. Annals of internal medicine, 108(1), 80-87.

Roll, I., & Wylie, R. 2016. Evolution and revolution in artificial intelligence in education. International journal of artificial intelligence in education, 26, 582-599.